

## Buffer Connections



Items required for the buffer

1 x Mellow LLL Buffer

2 x JST PHR-2

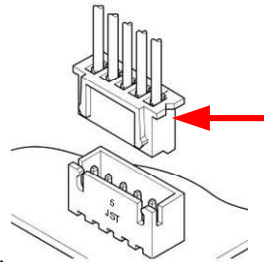
1 x JST XH 3-Way

1 x JST XH 4-Way

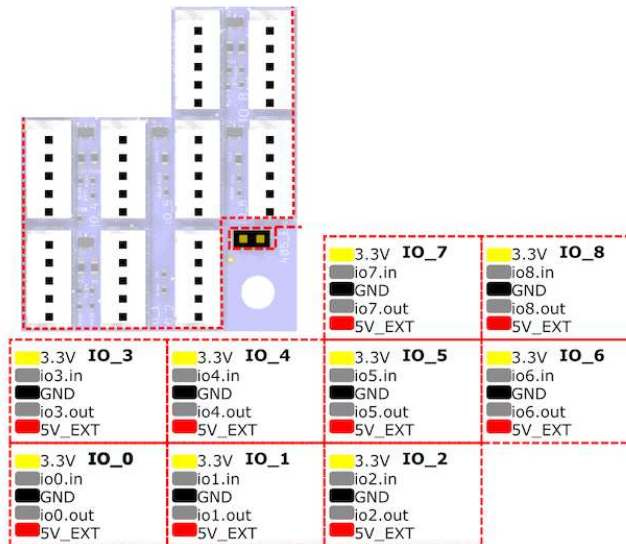
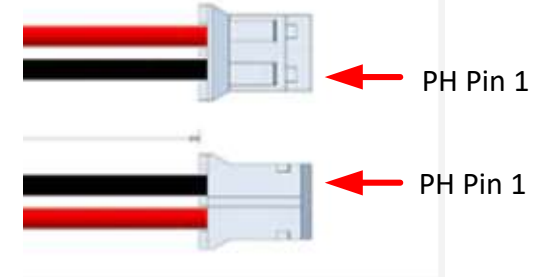
Control Board connections required

3 inputs,, Filament Detect , Button control Load/unload

2 outputs, control buffer load/unload



XH Notch Pin 1



Note:- Twist the supply wires together

Example of connections to a Duet 6HC

A1)12/24

A2)0V

A3)Filament Detect

A4)N/C

B1)0V signal

B2)Retract status

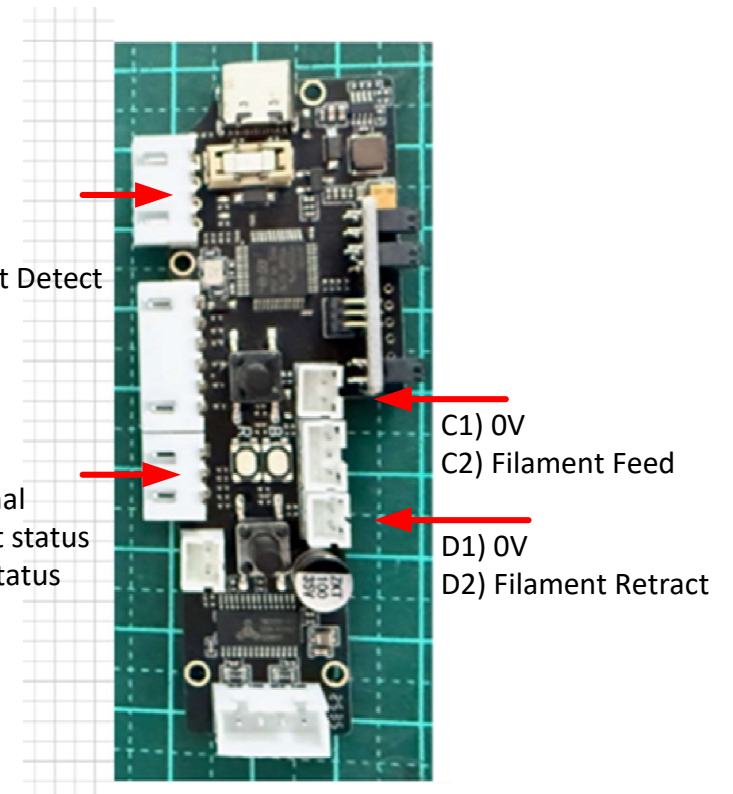
B3)Feed Status

C1) 0V

C2) Filament Feed

D1) 0V

D2) Filament Retract



| Con/Pin | From Buffer      | Operation               | wire Col/Size | TO                     |
|---------|------------------|-------------------------|---------------|------------------------|
| A1      | 12/24V           | Positive supply         | Red/22        | Positive 12/24V Supply |
| A2      | 0V               | 0V supply               | Black/22      | Negative DC supply     |
| A3      | Filament Detect  | Filament Detect Status  | White/22      | IO_8_in                |
| B1      | 0V signal        | Gnd Signal              | Black/22      | IO_7_GND               |
| B2      | Retract Status   | Filament Retract Signal | Blue/22       | IO_6_in                |
| B3      | Feed Status      | Filament Feed Signal    | Yellow/22     | IO_7_in                |
| C2      | Filament Feed    | Filament Buffer Feed    | White /24     | IO_7_out               |
| D2      | Filament Retract | Filament Buffer Retract | Green/24      | IO_8_out               |

JST XH crimp wire size 22 > 26AWG  
JST PH crimp wire size 24 > 30AWG  
Stripping Length: 1.9 – 2.5 mm.

## Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

| Buffer Pin      | Trigger Signal | Action Performed                      | Duration                              |
|-----------------|----------------|---------------------------------------|---------------------------------------|
| Feed Control    | Low Level      | Buffer continuously feeds filament    | Executes while the signal is held low |
| Retract Control | Low Level      | Buffer continuously retracts filament | Executes while the signal is held low |
|                 |                |                                       |                                       |

| Button         | Operation    | Signal Output (Buffer Pin)            | Signal Type | Duration                                            |
|----------------|--------------|---------------------------------------|-------------|-----------------------------------------------------|
| Feed Button    | Click        | FEED pin outputs a high-level pulse   | High Level  | Automatically returns to low level after 3 seconds  |
| Feed Button    | Press & Hold | Continuous feeding                    | High Level  | Until the button is released                        |
| Retract Button | Click        | RETRACT pin outputs a low-level pulse | Low Level   | Automatically returns to high level after 3 seconds |
| Retract Button | Press & Hold | Continuous retraction                 | Low Level   | Until the button is released                        |

Note: The buffer stops the action when the signal returns to a high level.

## Working Operation Buffer Buttons

**Click retract button** , will read the current filament loaded into DWC, heat the Hotend to the correct temperature and retract the filament .

**Click the Load button** , This will just load the filament until the filament stops automatically,

**To Cancel filament feeding press retract button for >1 second**

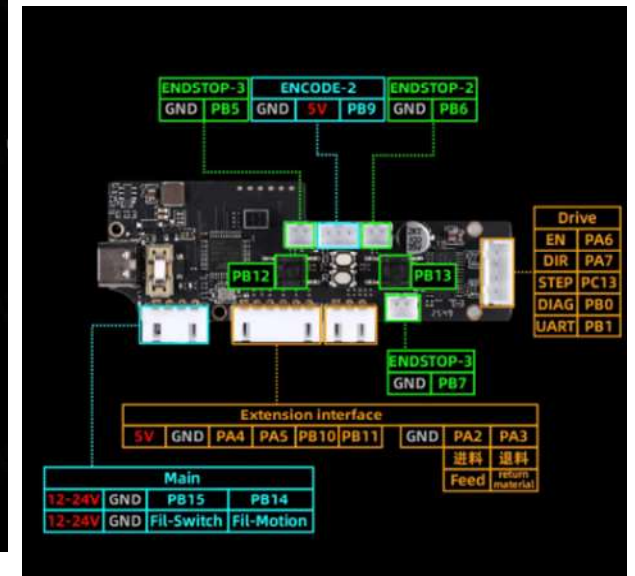
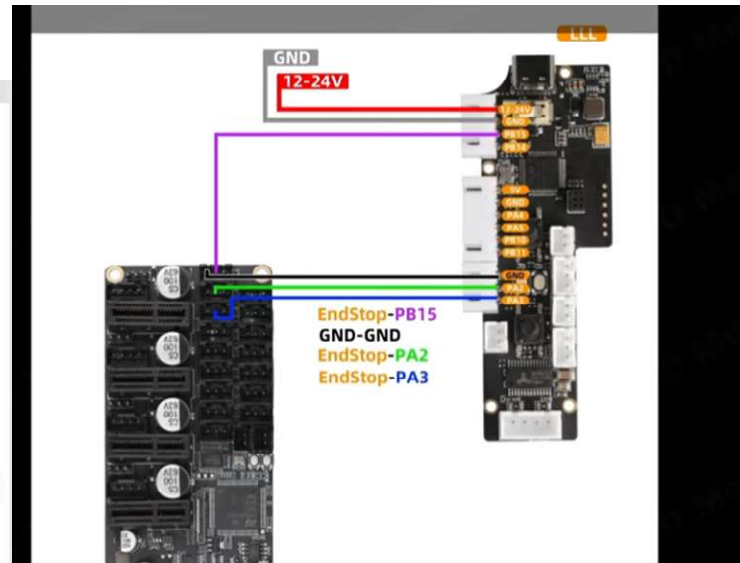
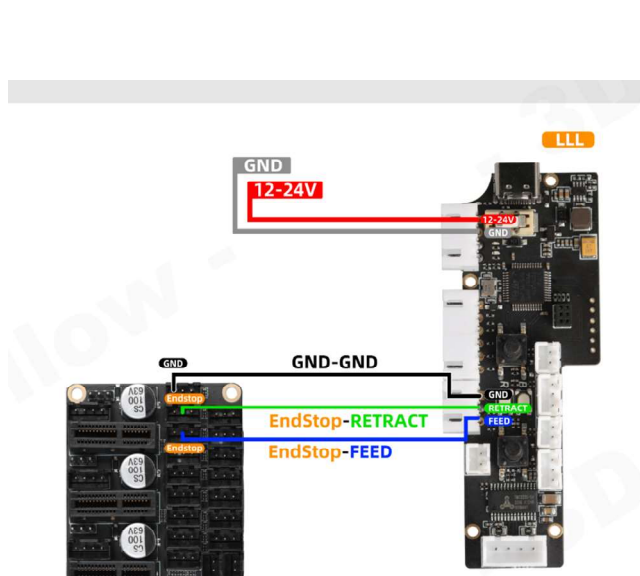
**Press and hold the Retract button , Will just force the buffer to retract the filament , make sure the filament is not still in the extruder!**

**Press and hold the Load button, will force the filament to feed until the filament starts skipping in the buffer, not recommended!**

### How to up date Firmware

<https://mellow.klipper.cn/en/docs/ProductDoc/ExtensionBoard/fly-buffer-plus/update>

<https://drive.google.com/drive/folders/1wjAlmvND2FCBo72GI6J4LZEO3XSbTHVE>



## Signal Output Description #

| Button                   | Operation  | Signal Output (Buffer Pin)            | Signal Type | Duration                                            |
|--------------------------|------------|---------------------------------------|-------------|-----------------------------------------------------|
| Feed Button (FEED)       | Click      | FEED pin outputs a high-level pulse   | High Level  | Automatically returns to low level after 3 seconds  |
| Feed Button (FEED)       | Long Press | Continuous feeding                    | High Level  | Until the button is released                        |
| Retract Button (RETRACT) | Click      | RETRACT pin outputs a low-level pulse | Low Level   | Automatically returns to high level after 3 seconds |
| Retract Button (RETRACT) | Long Press | Continuous retraction                 | Low Level   | Until the button is released                        |

## Working Principle

When the mainboard's GPIO pin outputs a low-level signal, the buffer detects this signal and performs the corresponding action:

| Buffer Pin | Trigger Signal | Action Performed                      | Duration                              |
|------------|----------------|---------------------------------------|---------------------------------------|
| PB5        | Low Level      | Buffer continuously feeds filament    | Executes while the signal is held low |
| PB6        | Low Level      | Buffer continuously retracts filament | Executes while the signal is held low |

Note: The buffer stops the action when the signal returns to a high level.